

Conceptual Questions Set 2.

1. The term "sex-linked" is most often used to describe genes on the X chromosome. Why is it rare to talk about "Y-linked" traits in humans, and what unique inheritance pattern would a rare Y-linked trait exhibit? Explain why this pattern is fundamentally different from X-linked inheritance.
2. A man with an X-linked dominant disorder has children with a woman who does not have the disorder. Predict the potential outcomes for their sons and daughters. Then, explain the biological reasoning behind why the pattern of inheritance for X-linked dominant conditions differs so markedly from that of X-linked recessive conditions.
3. In pedigree analysis of X-linked recessive traits, a common observation is that affected males are often related through unaffected females. Explain, at a molecular and cellular level, how a female can be "unaffected" yet still be the crucial link in transmitting the disorder across generations. What does this reveal about the relationship between genotype and phenotype in females for these traits?
4. The process of X-chromosome inactivation (X-inactivation) is often described as "random" in mammals. However, the resulting pattern of gene expression in the adult is not a uniform 50:50 mosaic. Using the concept of the cell lineage during embryonic development, explain how a random event at the cellular level can lead to large, clonal patches of cells expressing the same X chromosome in the adult (e.g., in calico cats). What does this imply about the timing of X-inactivation in development?
5. The phenomenon of "dosage compensation" implies a need for balance. Compare and contrast the mechanisms used to achieve dosage compensation in mammals (X-inactivation) with the mechanism used in fruit flies (*Drosophila*). Why is X-inactivation a "binary" solution (one X off) while *Drosophila* uses a "rheostat" solution (adjusting X activity)? What does this suggest about the different evolutionary paths to solving the same core problem?
6. Consider a woman who is heterozygous for an X-linked gene that is essential for the proper development of sweat glands. As a result of X-inactivation, some of her skin cells produce sweat glands and others do not, leading to a patchy condition where some areas of her skin cannot sweat. What would the phenotype be in a male with the same mutant allele? How does this comparison between the male and female phenotypes powerfully illustrate the consequences of both X-inactivation and its random nature?
7. Two genes are potentially "linked" because they are on the same chromosome. However, genes on same chromosome are not always inherited together. What is the meiotic event that can disrupt the linkage between two genes on the same chromosome? Explain how the physical distance between two genes on a chromosome influences the frequency of this event and is therefore the basis for constructing a linkage map.
8. A linkage map is a "genetic" map, not a physical map of DNA base pairs. Explain the distinction. If you have two pairs of genes, (A and B) and (C and D), and both pairs show a recombination frequency of 10%, can you conclude that the physical distance in base pairs between A and B is the same as that between C and D? Why or why not? What biological factors could cause discrepancies between genetic map distance and physical distance?
9. You are studying three genes on a chromosome. Your two-factor crosses show that the recombination frequency between genes 1 and 2 is 5%, and between genes 2 and 3 is 8%. However, the recombination frequency between genes 1 and 3 is not 13%, but something less, say 12.5%. How can the direct measurement between the outermost genes be less

than the sum of the intervening distances? What does this "missing" fraction of recombinants represent, and how does it reinforce the idea that a genetic map is based on the probability of a crossover event occurring between two points?